

Amarm	50219	AgSp1.1	tr	rm1	1	PGTTPGTVTGPDKPVKFIVPNG	AFLTPGSI	P	32
Amarm	50219	AgSp1.1	tr	rm1	2				0
Amarm	50219	AgSp1.1	tr	rm1	3	PGTTPGTVTGPDKPTKFIVPLG	AFSTPGSI	P	32
Amarm	50219	AgSp1.1	tr	rm1	4				0
Amarm	50219	AgSp1.1	tr	rm2	1	PGTTPGTVTGPDGRPTKFIIPLGA	FTTPGSI	P	32
Amarm	50219	AgSp1.1	tr	rm3	1				0
Amarm	50219	AgSp1.1	tr	cr1		PGTTPGTVTGPDGQPVKFVVPQGA	FSTPGTI	P	32
Amarm	50219	AgSp1.1	tr	cr		PGTTPGTVTGS DGNPTKFIVPFGAF	TPGSI	P	32
Amarm	50219	AgSp1.1	tr	cr2		PGTTPGTVTGPDGQPVKFVVPQGA	FSTPGTI	P	32
Amarm	50219	AgSp1.1	tr	cr3		PGTTPGTVTGPDGQPVKFVVPQGA	FSTPGTI	P	32
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	1	PGTTPGTVTDS DGPVRFIVPNGA	FMTPGSI	P	32
Lcorn	18104	AgSp1.1	tr	rm1	1	PGTTPGTVTDS DGPVRFIVPNG	AFLTPGSI	P	32
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	2				0
Lcorn	18104	AgSp1.1	tr	rm1	2				0
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	3	PGTTPGTVTGQDGKPTKFIVPIGA	FTTPGSI	P	32
Lcorn	18104	AgSp1.1	tr	rm1	3	PGTTPGTVTGQDGKPTKFIVPLG	AFTTPGSI	P	32
Lcorn	18104	AgSp1.1	tr	rm1	4				0
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	4				0
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	5	PGTTPGTVTGPDKPTKFIVPTGA	FTTPGSI	P	32
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	6				0
Lcorn	18104	AgSp1.1	tr	rm2					0
Lcorn	51676	AgSp1.1Cfrag	tr	cr	1	PGTTPGTVTGP DGNPTKFLVPLGAF	STPGSI	P	32
Lcorn	58650	AgSp1.2	tr	cr	1				0
Lcorn	58650	AgSp1.2	tr	cr	2	PGTTPGTVTGP DGNPTKFLVPLGAF	STPGSI	P	32
Lcorn	51676	AgSp1.1Cfrag	tr	cr	2				0
Lcorn	51676	AgSp1.1Cfrag	tr	cr	3	PGTTPGTVTGP DGEVVKFVVPQGA	STPGTI	P	32
Lcorn	58650	AgSp1.2	tr	cr	3				0
Lcorn	58650	AgSp1.2	tr	cr	4	PGTTPGTVTGP DGEVVKFVVPQGA	STPGTI	P	32
Ncruc	58532and13532	AgSp1	tr	1		PGTTPGTVTDP DGNPTKFIVPSGAFTT	PGSI	P	32
Ncruc	58532and13532	AgSp1	tr	2					0
Ncruc	58532and13532	AgSp1	tr	3		PGTITGTVTGP DGQPVKFVVPQGA	PGTI	P	32
Mlaby	7048	AgSp1	tr	rm	3				0
Mlaby	7048	AgSp1	tr	rm2	1			PGSPF	5
Mlaby	7048	AgSp1	tr	cr	1	PGSTPGTVTGP DGQPIKFVVPQGA	FSTPGTI	P	32
Mhut	1271	AgSp1	tr	rm1	1	PGTTPGTVTGP DGKPVRFIVPNGA	FMTPGSI	P	32
Mhut	1271	AgSp1	tr	rm1	2				0
Mhut	1271	AgSp1	tr	rm1	3	PGTTPGTVTGS DGRPTKFIVPIGA	FTTPGSI	P	32
Mhut	1271	AgSp1	tr	rm1	4				0
Mhut	1271	AgSp1	tr	rm2					0
Mhut	1271	AgSp1	tr	cr	1				0
Mhut	1271	AgSp1	tr	cr	2	PGNTPGTVTGS DGKPTKFIVPLGAFA	TPGSI	I	32
Mhut	1271	AgSp1	tr	cr	3				0
Mhut	1271	AgSp1	tr	cr	4	PGTTPGTVTGP DGQPVTFIVPQGAFS	TPGTMP	P	32
Aarg	AgSp1	FC511	tr	rm1	1	PGTTPGTVTDP DGKPVRFIVPNGAF	MTPGSI	P	32
Aarg	AgSp1	FC511	tr	rm1	2				0
Aarg	AgSp1	FC511	tr	rm1	3	PGTTPGTVTGP DGKPTKFIVPLGAF	TPGSI	K	32
Aarg	AgSp1	FC511	tr	rm1	4				0
Aarg	AgSp1	FC511	tr	rm1	5	PGTTPGTVTGS DGKPTKFIVPRGAF	TPGSI	P	32
Aarg	AgSp1	FC511	tr	rm1	6				0
Aarg	AgSp1	FC511	tr	rm2	1				0
Aarg	AgSp1	FC511	tr	rm2	2	PGTTPGTVTGP DGKPKKFVVPQGA	FTTPGSV	P	32
Aarg	AgSp1	FC511	tr	rm2	3				0
Atri	AgSp1	genomic	tr	rm1	1	PGTTPGTVTGP DGQLVRFIVPNGAF	MTPGSI	P	32
Atri	AgSp1	genomic	tr	rm1	2				0
Atri	AgSp1	genomic	tr	rm1	3	PGTTPGTVTGP DGKPTKFIVPLGAF	TPGSI	Q	32
Atri	AgSp1	genomic	tr	rm1	4				0
Atri	AgSp1	genomic	tr	rm1	5	PGTTPGTVTGS DGKPTRFIVPTGAF	TPGSI	P	32
Atri	AgSp1	genomic	tr	rm1	6				0
Atri	AgSp1	genomic	tr	cr		PGTTPGTVTGP DGNPTKFIVPLGAF	TPGSI	P	32
Msag	19329	AgSp1.1	tr	rm1		PGTTPGTVTGP DGNPNKFILPNGA	FTPGTI	P	32
Msag	18528	AgSp1.2	tr	rm	1	PGTTPGTVTGP DGKPKKFIVPN	MTGSI	P	32

[illegible]

Amarm	50219	AgSp1.1	tr	rm1	1	PQPQTT.....	TTKI..EDPHG..	KPIL..LPAG.....	45
Amarm	50219	AgSp1.1	tr	rm1	2	PQPQTT.....	TTKI..EDPHG..	KPIL..LPAG.....	52
Amarm	50219	AgSp1.1	tr	rm1	3	PQPQTT.....	TTKI..EDPHG..	KPIL..LPAG.....	45
Amarm	50219	AgSp1.1	tr	rm1	4	PSQT.....	PTPV..TGPSGQ..	PIK..LPAG.....	56
Amarm	50219	AgSp1.1	tr	rm2	1	TT.....	TTPI..PGPDG..	EPLS..VPAG.....	97
Amarm	50219	AgSp1.1	tr	rm3	1	PQSV.....	RKPDG..	QPIE..VPAG.....	50
Amarm	50219	AgSp1.1	tr	cr1	TT.....	TTPI..PGPGP..	EPLQ..VPAG.....	97	
Amarm	50219	AgSp1.1	tr	cr	VP.TT.....	TTPI..PGPGP..	EPLQ..VPAG.....	97	
Amarm	50219	AgSp1.1	tr	cr2	TT.....	TTPI..PGPGP..	KPLQ..VPAG.....	97	
Amarm	50219	AgSp1.1	tr	cr3				45	
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	1				45
Lcorn	18104	AgSp1.1	tr	rm1	1	PQPQTT.....	PTKI..ENPHG..	KPIL..LPAG.....	52
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	2	PQPQTT.....	PTKI..ENPRG..	KPIL..LPAG.....	52
Lcorn	18104	AgSp1.1	tr	rm1	2				45
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	3				45
Lcorn	18104	AgSp1.1	tr	rm1	3				45
Lcorn	18104	AgSp1.1	tr	rm1	4	PSQT.....	PTPV..MGPNG..	KPIQ..IPAG.....	56
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	4	PSQT.....	PTPV..KGPSG..	KPIQ..IPAG.....	56
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	5				45
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	6				56
Lcorn	18104	AgSp1.1	tr	rm2					60
Lcorn	51676	AgSp1.1Cfrag	tr	cr	1				45
Lcorn	58650	AgSp1.2	tr	cr	1	NPTT.....	PTDV..RGPQG..	NPIL..VPAG.....	54
Lcorn	58650	AgSp1.2	tr	cr	2				45
Lcorn	51676	AgSp1.1Cfrag	tr	cr	2	AP.IT.....	KTPI..PGPGP..	ETLQ..VPAG.....	52
Lcorn	51676	AgSp1.1Cfrag	tr	cr	3				45
Lcorn	58650	AgSp1.2	tr	cr	3	AP.IT.....	TTPI..PGPGP..	EPLQ..VPAG.....	52
Lcorn	58650	AgSp1.2	tr	cr	4				45
Ncruc	58532and13532	AgSp1	tr	1					45
Ncruc	58532and13532	AgSp1	tr	2	VP.IT.....	TTPV..PGP.G..	APIS..VPAG.....	51	
Ncruc	58532and13532	AgSp1	tr	3					45
Mlaby	7048	AgSp1	tr	rm	3	GNP...LNPGAQPTPRVGGQPRNPMFPLGPGGQPRPQS.PGQQTTT.....	PEPI..PGPNG..	KPLQ..VPAG.....	123
Mlaby	7048	AgSp1	tr	rm2	1				48
Mlaby	7048	AgSp1	tr	cr	1				45
Mhut	1271	AgSp1	tr	rm1	1				45
Mhut	1271	AgSp1	tr	rm1	2	SPQPQTT.....	TTNI..EGPYG..	KPIL..LPAG.....	52
Mhut	1271	AgSp1	tr	rm1	3				45
Mhut	1271	AgSp1	tr	rm1	4	PYQT.....	PSPL..KGPKG..	KPIQ..IPAG.....	56
Mhut	1271	AgSp1	tr	rm2	GYPPYQMTPGGQPINLYGPQDP..TTHIGLGSQSKFKTTQPITKPY..	SWNY..KGPHG..	KPIH..VPAG.....	113	
Mhut	1271	AgSp1	tr	cr	1	QK.....	GPN..	NPIL..VPAG.....	46
Mhut	1271	AgSp1	tr	cr	2				45
Mhut	1271	AgSp1	tr	cr	3	FP.IT.....	ASPT..PGSEP..	EPLQ..VPAG.....	52
Mhut	1271	AgSp1	tr	cr	4				45
Aarg	AgSp1	FC511	tr	rm1	1				45
Aarg	AgSp1	FC511	tr	rm1	2	PEPPTT.....	TTKI..EGPRG..	KPIL..LPAG.....	52
Aarg	AgSp1	FC511	tr	rm1	3				45
Aarg	AgSp1	FC511	tr	rm1	4	PSPT.....	PTPV..TGTEG..	KPLQ..IPAG.....	56
Aarg	AgSp1	FC511	tr	rm1	5				45
Aarg	AgSp1	FC511	tr	rm1	6	TPVTT.....	PTPV..PGPKG..	RPIQ..VPAG.....	58
Aarg	AgSp1	FC511	tr	rm2	1	STTT.....	PTSI..SGPDG..	QPMQ..VPAG.....	59
Aarg	AgSp1	FC511	tr	rm2	2				45
Aarg	AgSp1	FC511	tr	rm2	3	.NP...LSPNRQT.....TPGSEG.PGFKFQT.....	PRPI..KGPKG..	KPIQ..IPAG.....	75
Atri	AgSp1	genomic	tr	rm1	1				45
Atri	AgSp1	genomic	tr	rm1	2	PQPPTT.....	TTKI..EGPQG..	KPIL..LPAG.....	52
Atri	AgSp1	genomic	tr	rm1	3				45
Atri	AgSp1	genomic	tr	rm1	4	PSPT.....	PTPV..TGTEG..	KPIR..IPAG.....	56
Atri	AgSp1	genomic	tr	rm1	5				45
Atri	AgSp1	genomic	tr	rm1	6	KPATT.....	PTPV..PGPKG..	RPIQ..VPAG.....	58
Atri	AgSp1	genomic	tr	cr	VP.TT.....	TTPI..PGPGQ..	PIQ..LPAGPGTTPGTVTGP	111	
Msag	19329	AgSp1.1	tr	rm1	NT.....	LSPV..TKPDG..	YPIQ..ILAGPGKTPGIVPVKY...	116	
Msag	18528	AgSp1.2	tr	rm	1				45

Amarm 50219 AgSp1.1 tr rm1 1		45
Amarm 50219 AgSp1.1 tr rm1 2		52
Amarm 50219 AgSp1.1 tr rm1 3		45
Amarm 50219 AgSp1.1 tr rm1 4		56
Amarm 50219 AgSp1.1 tr rm2 1		97
Amarm 50219 AgSp1.1 tr rm3 1		50
Amarm 50219 AgSp1.1 tr cr1		97
Amarm 50219 AgSp1.1 tr cr		97
Amarm 50219 AgSp1.1 tr cr2		97
Amarm 50219 AgSp1.1 tr cr3		45
Lcorn 51676 AgSp1.2Nfrag tr rm1 1		45
Lcorn 18104 AgSp1.1 tr rm1 1		45
Lcorn 51676 AgSp1.2Nfrag tr rm1 2		52
Lcorn 18104 AgSp1.1 tr rm1 2		52
Lcorn 51676 AgSp1.2Nfrag tr rm1 3		45
Lcorn 18104 AgSp1.1 tr rm1 3		45
Lcorn 18104 AgSp1.1 tr rm1 4		56
Lcorn 51676 AgSp1.2Nfrag tr rm1 4		56
Lcorn 51676 AgSp1.2Nfrag tr rm1 5		45
Lcorn 51676 AgSp1.2Nfrag tr rm1 6		56
Lcorn 18104 AgSp1.1 tr rm2		60
Lcorn 51676 AgSp1.1Cfrag tr cr 1		45
Lcorn 58650 AgSp1.2 tr cr 1		54
Lcorn 58650 AgSp1.2 tr cr 2		45
Lcorn 51676 AgSp1.1Cfrag tr cr 2		52
Lcorn 51676 AgSp1.1Cfrag tr cr 3		45
Lcorn 58650 AgSp1.2 tr cr 3		52
Lcorn 58650 AgSp1.2 tr cr 4		45
Ncruc 58532and13532 AgSp1 tr 1		45
Ncruc 58532and13532 AgSp1 tr 2		51
Ncruc 58532and13532 AgSp1 tr 3		45
Mlaby 7048 AgSp1 tr rm 3		123
Mlaby 7048 AgSp1 tr rm2 1		48
Mlaby 7048 AgSp1 tr cr 1		45
Mhut 1271 AgSp1 tr rm1 1		45
Mhut 1271 AgSp1 tr rm1 2		52
Mhut 1271 AgSp1 tr rm1 3		45
Mhut 1271 AgSp1 tr rm1 4		56
Mhut 1271 AgSp1 tr rm2		113
Mhut 1271 AgSp1 tr cr 1		46
Mhut 1271 AgSp1 tr cr 2		45
Mhut 1271 AgSp1 tr cr 3		52
Mhut 1271 AgSp1 tr cr 4		45
Aarg AgSp1 FC511 tr rm1 1		45
Aarg AgSp1 FC511 tr rm1 2		52
Aarg AgSp1 FC511 tr rm1 3		45
Aarg AgSp1 FC511 tr rm1 4		56
Aarg AgSp1 FC511 tr rm1 5		45
Aarg AgSp1 FC511 tr rm1 6		58
Aarg AgSp1 FC511 tr rm2 1		59
Aarg AgSp1 FC511 tr rm2 2		45
Aarg AgSp1 FC511 tr rm2 3		75
Atri AgSp1 genomic tr rm1 1		45
Atri AgSp1 genomic tr rm1 2		52
Atri AgSp1 genomic tr rm1 3		45
Atri AgSp1 genomic tr rm1 4		56
Atri AgSp1 genomic tr rm1 5		45
Atri AgSp1 genomic tr rm1 6		58
Atri AgSp1 genomic tr cr	VKFIVPQGAFVTPGTILG.....PDGKPIPVE.....PQG.....	141
Msaag 19329 AgSp1.1 tr rm1	...VIPKGAFSTTPGYITGPYG.....	134
Msaag 18528 AgSp1 2 tr rm 1		45

Amarm	50219	AgSp1.1	tr	rm1	1		45
Amarm	50219	AgSp1.1	tr	rm1	2		52
Amarm	50219	AgSp1.1	tr	rm1	3		45
Amarm	50219	AgSp1.1	tr	rm1	4		56
Amarm	50219	AgSp1.1	tr	rm2	1		97
Amarm	50219	AgSp1.1	tr	rm3	1		50
Amarm	50219	AgSp1.1	tr	cr1			97
Amarm	50219	AgSp1.1	tr	cr			97
Amarm	50219	AgSp1.1	tr	cr2			97
Amarm	50219	AgSp1.1	tr	cr3			45
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	1		45
Lcorn	18104	AgSp1.1	tr	rm1	1		45
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	2		52
Lcorn	18104	AgSp1.1	tr	rm1	2		52
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	3		45
Lcorn	18104	AgSp1.1	tr	rm1	3		45
Lcorn	18104	AgSp1.1	tr	rm1	4		56
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	4		56
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	5		45
Lcorn	51676	AgSp1.2Nfrag	tr	rm1	6		56
Lcorn	18104	AgSp1.1	tr	rm2			60
Lcorn	51676	AgSp1.1Cfrag	tr	cr	1		45
Lcorn	58650	AgSp1.2	tr	cr	1		54
Lcorn	58650	AgSp1.2	tr	cr	2		45
Lcorn	51676	AgSp1.1Cfrag	tr	cr	2		52
Lcorn	51676	AgSp1.1Cfrag	tr	cr	3		45
Lcorn	58650	AgSp1.2	tr	cr	3		52
Lcorn	58650	AgSp1.2	tr	cr	4		45
Ncruc	58532and13532	AgSp1	tr		1		45
Ncruc	58532and13532	AgSp1	tr		2		51
Ncruc	58532and13532	AgSp1	tr		3		45
Mlaby	7048	AgSp1	tr	rm	3		123
Mlaby	7048	AgSp1	tr	rm2	1		48
Mlaby	7048	AgSp1	tr	cr	1		45
Mhut	1271	AgSp1	tr	rm1	1		45
Mhut	1271	AgSp1	tr	rm1	2		52
Mhut	1271	AgSp1	tr	rm1	3		45
Mhut	1271	AgSp1	tr	rm1	4		56
Mhut	1271	AgSp1	tr	rm2			113
Mhut	1271	AgSp1	tr	cr	1		46
Mhut	1271	AgSp1	tr	cr	2		45
Mhut	1271	AgSp1	tr	cr	3		52
Mhut	1271	AgSp1	tr	cr	4		45
Aarg	AgSp1	FC511	tr	rm1	1		45
Aarg	AgSp1	FC511	tr	rm1	2		52
Aarg	AgSp1	FC511	tr	rm1	3		45
Aarg	AgSp1	FC511	tr	rm1	4		56
Aarg	AgSp1	FC511	tr	rm1	5		45
Aarg	AgSp1	FC511	tr	rm1	6		58
Aarg	AgSp1	FC511	tr	rm2	1		59
Aarg	AgSp1	FC511	tr	rm2	2		45
Aarg	AgSp1	FC511	tr	rm2	3		75
Atri	AgSp1	genomic	tr	rm1	1		45
Atri	AgSp1	genomic	tr	rm1	2		52
Atri	AgSp1	genomic	tr	rm1	3		45
Atri	AgSp1	genomic	tr	rm1	4		56
Atri	AgSp1	genomic	tr	rm1	5		45
Atri	AgSp1	genomic	tr	rm1	6		58
Atri	AgSp1	genomic	tr	cr			141
Msag	19329	AgSp1.1	tr	rm1			134
Msag	18528	AgSp1	2	tr	rm	1	45

 non-conserved
 similar
 $\geq 50\%$ conserved